

StatSkill AI — Localhost Setup & Deployment Guide

Step-by-Step Engineering Runbook for Development & Localhost Execution

Repository: https://github.com/tanupatel-code/SIH26101_Project.git

System Architecture: StatSkill AI is a statistical competency assessment and microdata analytics portal for MoSPI. The system pairs a high-performance **FastAPI (Python)** backend on port 8000 with a responsive **Vite + React 19** frontend on port 5173.

0 System Prerequisites

Tool / Runtime	Minimum Version	Recommended	Verification Command
Python	3.10+	3.12.x	python --version
Node.js & npm	Node 18+, npm 9+	Node 20+ LTS	node -v && npm -v
Git CLI	2.30+	Latest stable	git --version
Docker (Optional)	24.0+ with Compose v2	Docker Desktop	docker compose version

1 Clone the Repository from GitHub

Open your terminal (PowerShell, Command Prompt, or Bash/Zsh) and clone the repository:

TERMINAL / SHELL

ALL PLATFORMS

```
git clone https://github.com/tanupatel-code/SIH26101_Project.git
cd SIH26101_Project
```

Directory Architecture:

```
SIH26101_Project/
|-- backend/           # FastAPI Application, Services & SQLite/JSON Store
|   |-- main.py        # Central REST API application entry point
|   |-- requirements.txt # Python dependencies (FastAPI, pypdf, pytest, etc.)
|   |-- .env.example   # Backend environment template
|   |-- statskill.json # User profile, competencies & document repository
|   |-- data_sources.json # 16 Official MoSPI surveys, registers & indices
|   |-- tests/         # 92 Automated pytest test suites
|-- frontend/         # Vite + React 19 Frontend User Interface
|   |-- App.jsx        # Root application router & UI workflows
|   |-- package.json   # Node dependencies
|   |-- .env.example   # Frontend environment template
|   |-- vite.config.js # Vite build & development server configuration
|-- START_BACKEND.bat  # 1-Click Windows Backend Launcher
|-- START_FRONTEND.bat # 1-Click Windows Frontend Launcher
|-- docker-compose.yml # Multi-container orchestration definition
```

2 Configure Backend Environment (.env)

Create the backend environment file in `SIH26101_Project/backend/.env`:

WINDOWS POWERSHELL

POWERSHELL

```
Copy-Item backend\.env.example backend\.env
```

WINDOWS CMD / MACOS / LINUX

```
# Windows Command Prompt:
copy backend\.env.example backend\.env

# macOS / Linux (Bash/Zsh):
cp backend/.env.example backend/.env
```

Verify that `backend/.env` contains:

```
STATSKILL_ADMIN_KEY=dev-admin-key
STATSKILL_CORS_ORIGINS=http://localhost:5173,http://127.0.0.1:5173,http://localhost:80
GEMINI_API_KEY=
OPENAI_API_KEY=
```

Offline Operation: GEMINI_API_KEY and OPENAI_API_KEY are optional. When omitted, StatSkill AI uses its built-in MoSPI/NSSTA question generation algorithms and official course curricula offline without needing cloud API keys.

3 Set Up Python Environment & Install Backend Dependencies

A Python virtual environment isolates dependencies cleanly:

WINDOWS (POWERSHELL)

MACOS / LINUX (BASH/ZSH)

```
python -m venv .venv
.venv\Scripts\Activate.ps1
```

```
python3 -m venv .venv
source .venv/bin/activate
```

PowerShell Tip: If script execution is restricted, run: Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass

Upgrade pip and install the required dependencies:

TERMINAL / SHELL (FROM PROJECT ROOT)

```
python -m pip install --upgrade pip
pip install -r backend/requirements.txt
```

Installed Packages: FastAPI (0.115.14), Uvicorn, Pydantic v2, pypdf (5.0+) for PDF generation, python-multipart for uploads, and pytest.

4 Configure Frontend Environment (.env)

Create the frontend environment file in `SIH26101_Project/frontend/.env`:

WINDOWS POWERSHELL / CMD / BASH

```
# Windows PowerShell:
Copy-Item frontend\.env.example frontend\.env

# Windows CMD:
copy frontend\.env.example frontend\.env

# macOS / Linux:
cp frontend/.env.example frontend/.env
```

Verify that `frontend/.env` contains:

```
VITE_API_BASE_URL=http://localhost:8000
```

5 Install Frontend Dependencies & Verify Build

Navigate to the frontend directory, install npm packages, and verify production build:

TERMINAL / SHELL (FROM PROJECT ROOT)

```
cd frontend
npm install
npm run build
cd ..
```

✓ Output: built in ~2s with 0 errors.

6 Launch Application on Localhost

Start both backend and frontend servers in separate terminal tabs:

Terminal 1: Start Backend API (FastAPI on Port 8000)

TERMINAL 1 (FROM SIH26101_PROJECT ROOT)

```
# Ensure virtual environment is active (.venv)
cd backend
python -m uvicorn main:app --reload --host 0.0.0.0 --port 8000
```

Backend URL: <http://localhost:8000> · Interactive Swagger API Docs: <http://localhost:8000/docs>

Terminal 2: Start Frontend UI (Vite on Port 5173)

TERMINAL 2 (FROM SIH26101_PROJECT ROOT)

```
cd frontend
npm run dev
```

Frontend Portal URL: <http://localhost:5173>

Windows 1-Click Launchers: Alternatively, double-click `START_BACKEND.bat` and `START_FRONTEND.bat` in the project root folder.

7 Portal Access & Pre-Configured Test Accounts

Open your browser and navigate to: <http://localhost:5173/>

Log in using either the 1-click login shortcuts or manual credentials:

User Persona	Official Email	Password	1-Click UI Shortcut
Senior Statistical Officer (SSO) Ananya Verma, JSO / Cadre Officer	ananya.verma@demo.gov.in	Demo@12345	Click "Quick Fill Demo Officer (SSO)"
General Citizen / Public Learner Aarav Sharma / Open Learner Track	aarav.sharma@gmail.com	Demo@12345	Click "Quick Fill Citizen / General Public"
Custom New Registration Self-registered officer or student	Your custom email	Your custom password	Click "Sign Up / Register"

8 Verification & Validation Runbook

Run the 92-test automated suite to confirm system integrity:

TERMINAL / SHELL (FROM PROJECT ROOT)

```
python -m pytest backend/tests -v
```

✓ Result: **92 passed** (Components, Type Safety, Deployment Readiness, E2E Workflows).

Key Feature Smoke Tests:

Feature	Navigation Path	Expected Output
Document Vault	Sidebar → My Documents	8 distinct official MoSPI study resources across Survey Methodology, National Accounts, Price Indices, and GIS.
Document Download	Click download icon on any document	Downloads clean, openable PDF/DOCX material.
Official Certificate	Sidebar → Certificates → Click "Verify Credential"	Opens modal; clicking " Download Official Certificate " downloads valid PDF 1.4 file that opens in Acrobat and Chrome.
Data Sources	Sidebar → Data Sources	Displays 16 official statistical surveys with functional domain filters and live search.
Course Analytics	Sidebar → Analytics	Displays competency score radars, domain effort hours, and learning velocity.
iGOT Learning Path	Sidebar → Learning Path	Displays recommended modules with interactive quiz player and study workspace.

9 Alternative: 1-Command Docker Deployment

If you have Docker Desktop installed, deploy the entire stack with a single command:

TERMINAL / SHELL (FROM PROJECT ROOT)

```
docker compose up --build -d
```

Access frontend at **http://localhost:5173** or **http://localhost:80**, and backend at **http://localhost:8000**.

To stop: `docker compose down`

10 Troubleshooting & FAQs

Issue	Root Cause	Resolution
Port 8000 or 5173 in use	Previous process still bound to port.	Windows PowerShell: <code>Get-Process -Id (Get-NetTCPConnection - LocalPort 8000).OwningProcess Stop-Process -Force</code> Linux/macOS: <code>kill \$(lsof -t -i:8000)</code>
PowerShell execution policy	Restricted script execution in Windows.	Run: <code>Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass</code>
CORS error in browser	Frontend port mismatch.	Verify <code>STATSKILL_CORS_ORIGINS</code> in <code>backend/.env</code> includes <code>http://localhost:5173</code> .
ModuleNotFoundError	Virtual environment not activated.	Activate <code>.venv</code> first, then run <code>pip install -r backend/requirements.txt</code> .